

The QueryGraph stack

Shared data, shared meaning, capable agents

Sail · LakeCat · Pinax · TypeSec · MCP · Marciana · Grust



A company asks a question

“Where is our customer data, and what can an analytics agent do with it?”

- Business teams use different words for the same concept
- Systems represent similar records with different columns
- Agents need meaning, structure, and usable operations

A lakehouse in Sail

Sail executes reads over Iceberg tables and Parquet records.

Field ID	Customer fixture column	Physical type
1	id	long, required
2	tenant	string, optional
3	private_email	string, optional

Console step 1 · Inspect lakehouse & catalog

A catalog in LakeCat

The catalog identifies the physical table and the snapshot an operation will use.

- Table identity and schema
- Current snapshot and catalog state
- Planning and execution through the Sail adapter

One table, described physically by LakeCat and semantically by Pinax

A registry in Pinax

Standardize the shape of tables across a large company

Shared profile

Common foundation

Customers

Stable identities, typed fields, semantic terms

Employees

Common HR records and stewardship

Transactions

Consistent financial record structure

Console step 2 · Generate enterprise standards

Shared meaning in the ontology

Account number → Customer identifier → rows.id

- Stable concepts, definitions, and aliases
- Reviewed bindings to registered tables and field IDs
- One maintained vocabulary for agents and exports

Console steps 3–4 · Discover tables, then consult the ontology

Agents with TypeSec access

An identified agent discovers the tables and fields appropriate to its task.

- The analytics agent carries a TypeSec identity
- Pinax returns its usable table inventory
- Its requested operation carries a purpose and signature

Access is part of the normal workflow, alongside schema and meaning.

MCP is the agent's interface

Discover tools

- Search “account number” in Pinax
- Resolve the table and field
- Call `execute_pinax_scan`
- Receive rows and semantic metadata

Console step 5 · Discover & read through MCP

Plan and read the selected table

```
{"table":"rows","columns":["id"],  
  "purpose":"analytics","limit":10}
```

The plan identifies the current LakeCat snapshot.

Sail returns `[{"id":2}]`

Pinax's semantic exports preserve the selected field's meaning.

Console steps 6–7 · Inspect plan, read selected rows

Grust connects the data

Datasets → files → fields → concepts

Live demonstration

Result

**Dataset
descriptions**

Two Dataverse fixtures

Property graph

34 nodes and 33 edges stored through Sail

Verification

Dataset node read back from Sail

Provenance

OpenLineage appended to the lakehouse

Console step 8 · Build graph & run workflow

Marciana and semantic workflows

QueryGraph composes semantic and cognition capabilities into agent workflows.

- Navigator packages descriptions for discovery and reuse
- Specialists contribute domain summaries
- A supervisor coordinates and a synthesis agent combines
- TypeSec identities and lineage accompany the work

Console steps 9–10 · Semantic bundle and specialist workflow

Meaning travels with results

Surface

What an agent can reuse

Croissant

Record sets, fields, and bounded inline results

CDIF projection

Data elements linked to shared concepts

SKOS vocabulary

Labels, aliases, definitions, relationships

Lineage

The operations and inputs behind an answer

One central ontology feeds the semantic projections.

Run the whole-stack walkthrough

```
ssh -N -L 18081:127.0.0.1:18081 grust
```

```
# Open http://localhost:18081
```

Results stay on the left. Steps scroll on the right.

Click any component to see its role and a related operation.

Ten main steps · optional access checks · inspectable JSON

Pinax

The agentic entrypoint to the QueryGraph stack

Shared table shapes

Shared business meaning

Discoverable operations

firstpair.org/books/pinax/

querygraph.ai/announcing-pinax/

